

1. Automatic Control System for Thermal Comfort Based on Predicted Mean Vote and Energy Saving

Objective

This paper develops an automatic air-conditioning control system that regulates indoor thermal comfort using the Predicted Mean Vote (PMV) index instead of fixed temperature setpoints.

Problem Addressed

The main problem is that Fanger's PMV model is nonlinear and depends on multiple variables, which makes real-time inversion difficult for control purposes. Conventional fixed-temperature control is easier to implement, but it is often energy-inefficient and does not directly reflect human thermal comfort.

Proposed Method

The system combines three components:

1. **Wireless Sensor Network (WSN)** for real-time measurement of indoor conditions such as humidity, air velocity, and radiant temperature.
2. **Inverse PMV modeling** using two computational intelligence approaches:
 - Particle Swarm Optimization (PSO)
 - Adaptive Neuro-Fuzzy Inference System (ANFIS)
3. **Digital Self-Tuning Control (STC)** to regulate the air conditioner by adjusting the setpoint through an infrared emitter, without modifying the manufacturer's hardware.

The system assumes fixed occupant conditions of 1.1 met for metabolic rate and 0.5 clo for clothing insulation.

Key Results

PSO performed better than ANFIS for inverse PMV estimation:

- **RMSE:** 0.0075 for PSO versus 0.0335 for ANFIS
- **MAE:** 0.0051 for PSO versus 0.0264 for ANFIS

In experiments, the STC-based system maintained PMV within the ASHRAE Class A comfort range of ± 0.5 and achieved **energy savings of 30.1% to 39.4%** compared with fixed-point cooling at 23°C and 25°C.

Limitations

The system depends heavily on a dense WSN. Sensor failure or communication latency in the ZigBee network may affect control stability. In addition, the use of fixed clothing and activity assumptions creates a gap between modeled and actual comfort conditions.

2. CMOS Compatible MEMS Air Velocity Sensor With Improved Sensitivity and Linearity for Human Thermal Comfort Sensing Applications

Objective

This paper designs a CMOS-compatible MEMS air velocity sensor for low-speed indoor airflow measurement, targeting thermal comfort applications where accurate air velocity data are needed for PMV calculation.

Problem Addressed

Low-magnitude air velocity sensing is difficult because conventional hot-wire and calorimetric sensors often suffer from parasitic heat loss to the substrate, which reduces sensitivity and increases power consumption. Existing designs also struggle to meet the accuracy requirements of thermal comfort standards in open indoor environments.

Proposed Method

The authors propose a **thermoresistive calorimetric MEMS sensor** that measures airflow through temperature asymmetry around a micro-heater. A Wheatstone bridge converts the thermal gradient into an output voltage. The design is optimized for the 0 to 1 m/s range relevant to indoor comfort sensing.

The sensor is compared with earlier CMOS MEMS hot-wire and single-pair detector designs.

Key Results

The proposed sensor achieved:

- **Maximum sensitivity:** 340.2 mV/(m/s)
- **Normalized sensitivity:** 354.37 μ V/(m/s)/mW
- **Linearity:** coefficient of determination of 0.9996
- **Accuracy:** ± 0.0236 m/s

This accuracy is better than the ± 0.05 m/s threshold required for indoor thermal comfort measurement standards.

The optimized design improved performance by nearly **55%** relative to baseline single-pair detector configurations.

Limitations

The design favors linearity over maximum sensitivity by selecting a heater-to-detector spacing of 63 μ m instead of the 130 to 150 μ m range that gave higher simulated sensitivity. Also, completely released membranes remain structurally fragile and may not be robust in harsher environments.

A further limitation is that the sensor measures airflow in one direction only, so it must be oriented correctly.

3. iTCM: Toward Learning-Based Thermal Comfort Modeling via Pervasive Sensing for Smart Buildings

Objective

This paper proposes a learning-based thermal comfort modeling framework that combines environmental sensing with physiological data from occupants.

Problem Addressed

Traditional PMV-based models often fail to capture individual differences in comfort because they rely mainly on environmental variables and simplified assumptions about occupants. This leads to prediction bias and limited personalization.

Proposed Method

The paper introduces **iTCM**, a pervasive sensing framework that integrates:

1. **Ambient sensing** through a wireless sensor network
2. **Physiological and behavioral data** collected through a mobile application and wearable devices
3. **Machine learning models**, including deep learning, to predict individual comfort more accurately than PMV

The paper also explores a **deep reinforcement learning** application to estimate the economic value of thermal comfort and assess whether incentive-based strategies can influence occupant behavior to reduce electricity use.

Key Results

The proposed ITCNN model outperformed the traditional PMV model by:

- **13.1% average improvement**
- **Up to 17.8% improvement** in predictive accuracy

The work also showed, through simulation, that incentive-based mechanisms could potentially alter user behavior in ways that lower electricity costs.

Limitations

The framework requires continuous wearable connectivity, mobile interaction, and substantial data processing infrastructure. This increases deployment complexity and data overhead.

Another limitation is that the reinforcement learning energy-saving component was not fully validated in a real building with real occupant behavior. As a result, there is still uncertainty about actual occupant compliance with incentive schemes.

4. Learning-Based Hierarchical Distributed HVAC Scheduling With Operational Constraints

Objective

This paper develops a hierarchical distributed HVAC scheduling method for large multizone buildings, with explicit attention to operational constraints such as chiller capacity and ventilation requirements.

Problem Addressed

Centralized Model Predictive Control becomes computationally difficult in buildings with hundreds of zones. At the same time, many distributed control methods simplify the problem too much and ignore important plant-side constraints such as chiller bottlenecks and fresh air requirements.

Proposed Method

The paper proposes a **hierarchical distributed optimization framework** in which:

- Individual zones act as local decision units with their own Model Predictive Controllers
- A **Central Scheduler** allocates cooling resources through a token-based mechanism
- The upper-level problem is formulated as a **quadratically constrained quadratic program**
- Local learning modules estimate thermal disturbances such as stray heat and occupancy-related gains

This architecture extends earlier token-based scheduling approaches by including chiller-side and ventilation constraints.

Key Results

The proposed learning-based token scheduling approach achieved:

- **14% to 22% energy savings** in a ten-zone building compared with decentralized baselines
- Successful validation in a **500-zone simulation**
- Less than **3% performance gap** relative to the centralized optimum, despite much lower computational burden

Limitations

The token abstraction introduces some suboptimality relative to a fully centralized solution. Performance also depends on the quality of the Moving Horizon Estimation process used to estimate thermal disturbances. If rapid occupancy changes are not captured accurately, short-term comfort violations may occur.

Another important limitation is that the framework focuses mainly on **sensible cooling** and does not fully address humidity or latent heat loads.

5. Model Predictive Climate Control of a Swiss Office Building: Implementation, Results, and Cost-Benefit Analysis

Objective

This paper evaluates the real-world implementation of Model Predictive Control (MPC) in a large Swiss office building and examines both technical performance and economic viability.

Problem Addressed

Traditional Rule-Based Control (RBC) is unable to exploit building thermal inertia or coordinate multiple interacting systems efficiently. However, deploying MPC in real buildings remains difficult because it requires accurate building models, integration effort, and high upfront cost.

Proposed Method

The authors implement a **supervisory MPC framework** based on a bilinear, physics-based building model. The controller optimizes control trajectories over a prediction horizon while respecting comfort constraints.

The building includes:

- Thermally Activated Building Systems (TABS)
- Air handling units for faster heating and cooling response
- Automated blinds for solar gain management

Performance is compared with the original rule-based strategy, and EnergyPlus simulations are used to support whole-year analysis.

Key Results

The paper shows that MPC improved both comfort compliance and energy efficiency relative to the baseline RBC system.

The measured baseline building consumption used for reference was:

- **46 kWh/m² for heat**
- **33 kWh/m² for electricity**

Although the summary does not provide a single headline percentage reduction, the reported outcome is that MPC performed better than RBC in both thermal regulation and energy management.

Limitations

The largest limitation is economic rather than technical. The cost of model development, software engineering, and system integration is high, and those costs may exceed operational savings in many projects.

Model mismatch also remains an issue because building-specific physics-based models are labor-intensive to develop and maintain.

6. Stochastic Model Predictive Control for Building Climate Control

Objective

This paper develops a Stochastic Model Predictive Control (SMPC) approach for building climate control under uncertainty, especially uncertainty in weather forecasts and internal gains.

Problem Addressed

Deterministic MPC assumes forecasts are accurate and therefore can lead to frequent comfort violations when disturbances differ from predictions. Robust MPC avoids this by planning for worst-case conditions, but it can become overly conservative and waste energy.

Proposed Method

The authors formulate a **stochastic optimization problem with chance constraints**. To make the problem tractable, they use **Affine Disturbance Feedback**, which expresses control inputs as affine functions of past disturbances.

The method is evaluated using building models from the Swiss OptiControl project and is compared with:

- Deterministic MPC under perfect information
- Deterministic MPC with imperfect forecasts

Key Results

The SMPC method achieved a better trade-off between energy use and comfort than deterministic MPC under forecast uncertainty.

Key findings include:

- Deterministic MPC with imperfect forecasts caused comfort violations above **20% of occupancy time** in some cases
- SMPC kept violations within specified **5% or 10% chance-constraint bounds**
- Energy performance remained within roughly **5% to 10%** of the perfect-foresight benchmark

Limitations

The approach increases online computational complexity because it replaces a simpler linear optimization problem with a larger quadratic one involving disturbance covariance terms. The authors report that this is still feasible within a **15-minute control interval**, but the burden is higher.

Another limitation is the assumption that forecast errors follow a Gaussian distribution. Real disturbances may have heavy tails or non-Gaussian behavior, which could reduce the reliability of chance-constraint guarantees.

The method also relies on linear approximations and does not fully represent nonlinear building dynamics, moisture transport, or nonlinear equipment behavior.